FINAL PROJECT / MACHINE LEARNING

Predicting property prices in Azerbaijan-Overfitters Team

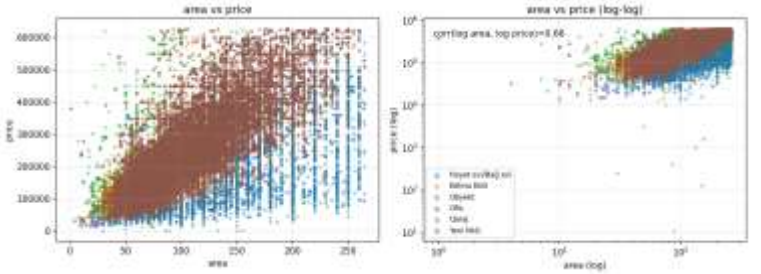
From-scratch decision trees and Pegasos SVM on bina.az listings

TASK A / LOG-PRICE REGRESSION

TASK B / PRICE-TIER CLASSIFICATION

93,150 cleaned listings | 28 features

Teymur Alasgarov | Tofiq Heydarov | Parvin Salahov



LISTING AREA AND PRICE

Turning listings into model inputs

RAW LISTINGS
100,775
before duplicate handling

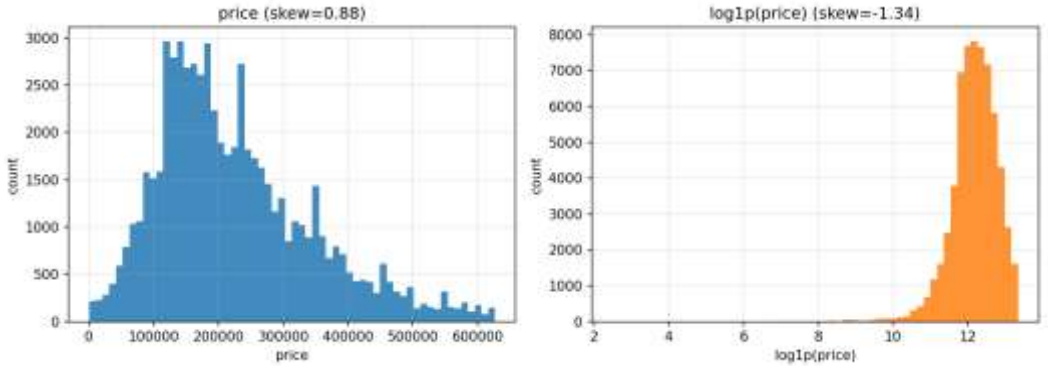
ANALYTICAL ROWS
93,150
after filtering and cleaning

INPUT FEATURES
28
after encoding

MEDIAN PRICE
200,000 AZN
training split

WHAT WE DID

- Handled duplicate listings; 100,773 rows remained.
- Prepared area, land area, floor, rooms, city and category features.
- Removed IDs, URLs, scrape-time fields and price-derived columns such as unit_price and total_price.
- Price was treated in source units (reported as AZN); no currency-conversion step is documented.



Log(area) vs log(price): 0.658 | Rooms vs log(price): 0.400

The log transform reduces the raw-price skew in the analyzed training data.

Two tasks, one shared dataset

TASK A / REGRESSION

Predict log(1 + price)

Metrics: RMSE | MAE | R-squared

TASK B / CLASSIFICATION

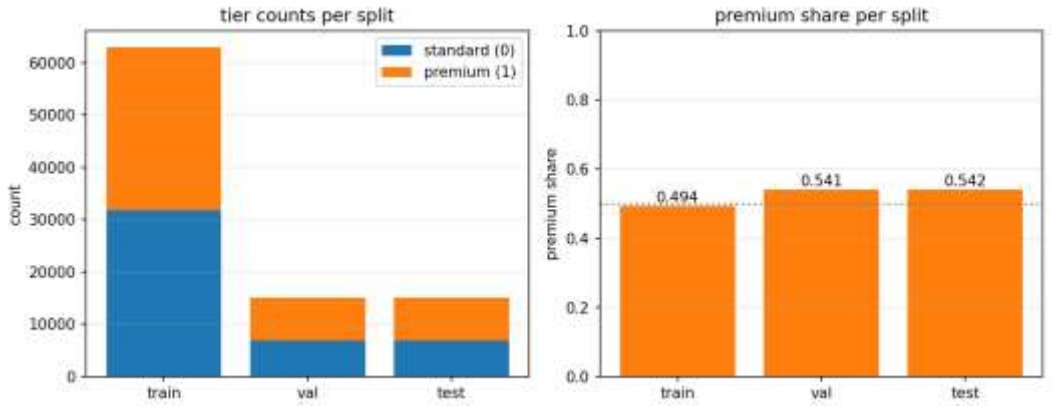
Premium vs standard

Premium = above training-price median

Metrics: precision | recall | F1 | AUC

SPLITS AND TIER BALANCE

Train 62,922 Validation 15,114 Test 15,114



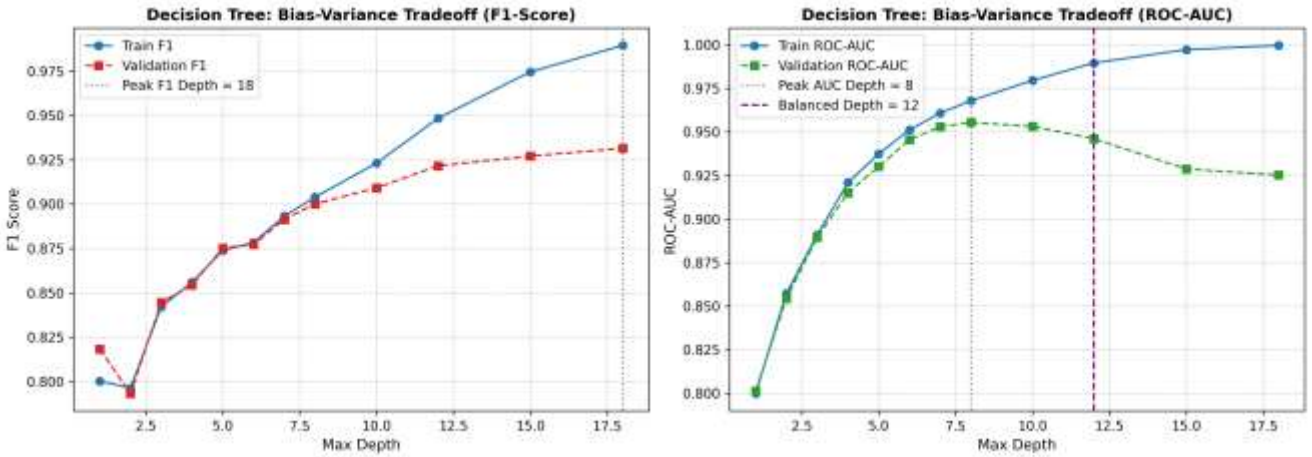
Premium share: 49.4% train | 54.1% validation | 54.2% test

The split used price quartile bins; it was not stratified directly on the tier label.

Decision tree: choose splits that reduce impurity

IMPLEMENTATION

- Recursive NumPy tree with numeric threshold search.
- Regression splits reduce variance / MSE; leaves predict the mean.
- Classification splits use Gini or entropy; leaves report class frequencies.
- Stopping controls: max depth, minimum split/leaf samples, and minimum impurity decrease.
- Tree hyperparameters chosen with validation data.



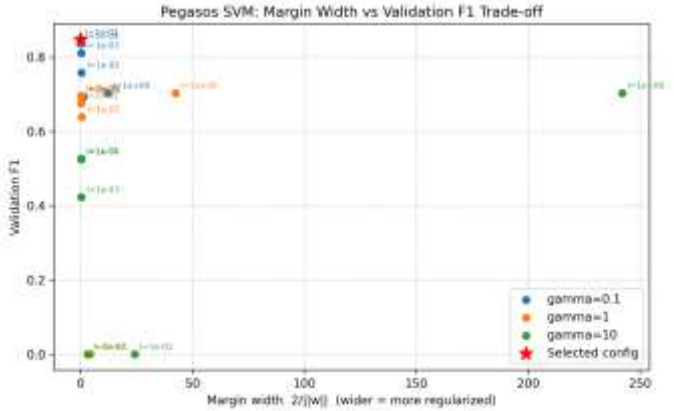
At deep settings, training AUC approaches 1.00 while validation AUC falls: a clear overfitting signal.

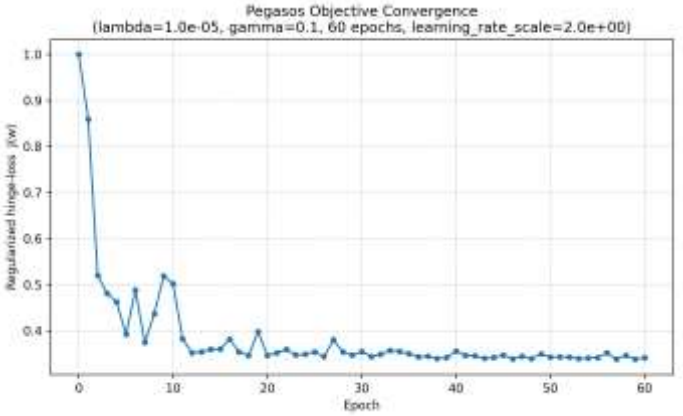
Pegasos SVM: hinge loss plus random features

THE MODEL

J(w,b) = 0.5*lambda*||w||^2 + mean(max(0, 1 - y*f(x)))

- Stochastic sub-gradient updates; fixed random seed.
- Inputs standardized from training statistics.
- 256 Random Fourier Features approximate an RBF kernel.
- Best validation grid: lambda=1e-5, gamma=0.1 (F1=0.8469; AUC=0.9073). Margin = 2/||w|| in RFF space.

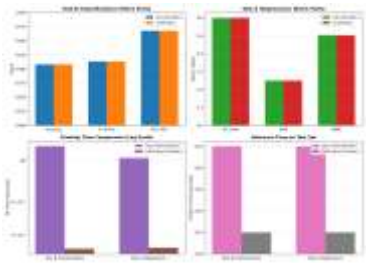




Held-out results: trees are closely matched

Test-set metrics from the full pipeline run (15,114 listings)

Model	Task A RMSE	Task B accuracy	Task B F1	Task B ROC-AUC
Scratch decision tree	0.5036	0.9079	0.9131	0.9675
scikit-learn decision tree	0.5028	0.9078	0.9131	0.9674
Pegasos RFF (main run)	N/A	0.8511	0.8612	0.9282
scikit-learn SGD hinge	N/A	0.8478	0.8578	0.9198



The scratch tree is within rounding distance of the sklearn tree on this split. Pegasos outperforms the linear SGD baseline on F1/AUC in this main run.

Separate SVM benchmark below uses train + validation for fitting after model selection; its results are reported on Slide 7.

Benchmark context and where errors remain

SVM BENCHMARK / 5-FOLD TASK B F1

Model	F1 mean +/- std	ROC-AUC mean +/- std
Pegasos RFF	0.8397 +/- 0.0164	0.9223 +/- 0.0070
RBF SVC	0.8874 +/- 0.0052	0.9570 +/- 0.0030
SGD hinge	0.8297 +/- 0.0070	0.9022 +/- 0.0063

SVC performed best in this separate benchmark.

PEGASOS TEST CONFUSION MATRIX

	Pred. standard	Pred. premium
Actual standard	5,989	939
Actual premium	1,545	6,641

Pegasos misses more premium listings than it falsely labels as premium.

LIMITATIONS TO DISCUSS

- Land and commercial listings have much higher regression errors than common apartment types.
- Sparse categories and regions have fewer examples; their scores are less stable.
- RFF approximates, but does not exactly reproduce, the RBF kernel.

What we learned - and what comes next

TEAM

Teymur Alasgarov ~ 40 %

Tofiq Heydarov ~ 27%

Parvin Salahov ~ 33 %

Main takeaway

- Our from-scratch tree performed close to scikit-learn: Task B F1 was **0.9131**; Task A RMSE was **0.5036** versus **0.5026**.
  Pegasos with Random Fourier Features outperformed the linear SGD baseline, while the exact RBF SVC achieved higher scores.
  What comes next
  Explore whether more Random Fourier Features improve the SVM's accuracy.
  Add more data or location features to better handle rare property types and districts.
  Thank you — Questions?